Bihar Engineering University, Patna

B.Tech 7th Semester Examination, 2024

Course : B.Tech | *Time: 03 Hours*

Code : 100708 | *Subject: Biology for Engineers* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.

*(ii) There are **NINE** questions in this paper.*

*(iii) Attempt **FIVE** questions in all.*

*(iv) Question No. **1** is compulsory.*

Q.1 Choose the correct answer of the following (any seven question only): **[2 x 7 = 14]**

(a) Which vitamin helps the body absorb calcium?
(i) Vitamin A (ii) Vitamin B
(iii) Vitamin C (iv) Vitamin D

(b) _____ discovered Brownian motion.
(i) Julius Robert Mayer (ii) Robert Brown
(iii) Justin Bieber (iv) Jennifer Doudna and Richard Roblin

(c) _____ is the most common prokaryotic model organism in Science.
(i) Arabidopsis thaliana (ii) C. elegans
(iii) Escherichia coli (iv) Salmonella typhi

(d) Which of the following is not present in animal cell?
(i) Ribosome (ii) Cytoplasm
(iii) Cell walls (iv) Mitochondria

(e) The term enzyme was given by?
(i) Traube (ii) F.W Kunne
(iii) Willstatter (iv) Louis Pasteur

(f) The term biology was introduced by
(i) John ray (ii) Aristotle
(iii) Lamarck and Treviranus (iv) Darwin

(g) Normal human blood pressure is?
(i) 110/100 (ii) 140/60
(iii) 120/70 (iv) 120/80

(h) What is a non-reducing carbohydrate?
(i) Sucrose (ii) Fructose
(iii) Glucose (iv) Maltose

(i) Inulin is characterized by _____
(i) Vitamin (ii) Polysaccharide
(iii) Lipid (iv) Protein

(j) Which enzyme catalyzes the hydrolysis of glucose?
(i) Trypsin (ii) Peptin
(iii) Glucosidase (iv) Alcalase

Q.2 (a) What is the difference between unicellular and multicellular reproduction? Explain with figures. **[7]**

(b) What do you understand by Thermodynamics? Explain with suitable examples referring to the original observation of Robert Brown **[7]**

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(b) What was the original observation of Robert Brown and Julius Mayor? [7]

Q.4 (a) What are the important applications of Biology for engineers? Explain in details. [7]
(b) Describe the Gene. How genetic material passes from parent to offspring. [7]

Q.5 (a) What do you understand by Monomers? Compare monomer with polymers with suitable example [7]
(b) What type of sugar is found in nucleotides? Explain the role of sugar nucleotides in biosynthesis. [7]

Q.6 (a) Explain the major functions of self-healing bio concrete. [4]
(b) What is Ecosystem? Explain the structure and function on Ecosystem. [10]

Q.7 What is genetic material of DNA? Explain molecular basis of coding and decoding genetic information. [14]

Q.8 What were the scientific advancements in the 18th century? Explain the examples of various biological observation that lead to major discoveries. [14]

Q.9 Write short notes on ***any two*** of the following:- **[7 x 2 = 14]**
(a) Brief note on cell.
(b) What do you understand about gene mapping?
(c) Define Mendel first law.
(d) Energy flow in an ecosystem

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